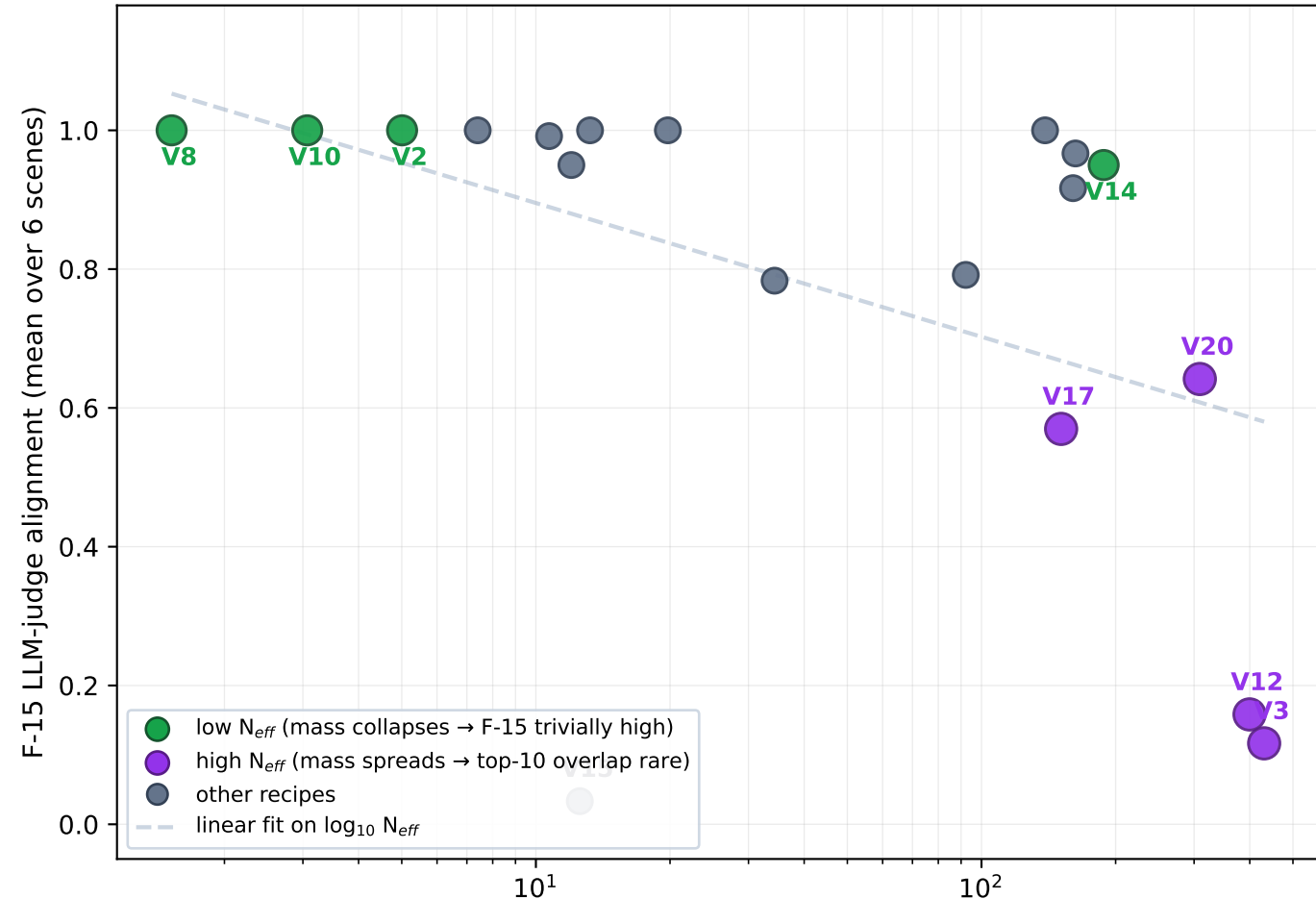


P5 mechanism — F-15 alignment tracks token-mass dispersion, not coherence
Spearman $\rho(\text{F-15}, N_{\text{eff}}) = -0.62$ (vs $\rho(\text{F-15}, |V|) = -0.57$; see companion figure)



Token-mass dispersion $N_{\text{eff}} = \exp H(\phi_k)$ (per-topic mean effective vocabulary, mean over 6 scenes, log scale)

$N_{\text{eff}} = \exp H(\phi_k)$ with ϕ_k the LDA topic-word distribution (mean over topics, then 6 scenes), computed from data/local/v_sweep/lda_fits/<scene>_<recipe>_uniform_Q8/phi.npy. F-15 from data/derived/v_sweep/f15_llm_alignment/. At the shared $|V|=1376$ alphabet, phi dispersion falls monotonically $V3 \approx 431 > V12 \approx 400 > V20 \approx 309$ as F-15 rises $0.12 < 0.16 < 0.64$; V14 ($|V|=1024$) has $N_{\text{eff}} \approx 188$ yet F-15 0.95: F-15 is a dispersion meter, refuting the pure $|V|$ reading.